NEST EDUCATION IT SCHOOL

2025-2026 school year

TETRIS

Demoday project report

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Ulaanbator city, 2026

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KEYWORDS, LIST OF ABBREVIATED WORDS

Arduino UNO

Arduino UNO is a small programmable smart control board designed for creating electronic projects and robots.

IoT

IoT stands for **“Internet of Things”** is a type that connects physical objects with software.

Joystick

A joystick is a stick-shaped input device that rotates on a base, used to control the movement of digital objects, play games, operate aircraft, remote controls, and industrial machinery.

LED

LED (Light Emitting Diode) is a long-lasting lighting technology that converts electric current into light.

PROJECT INTRODUCTION

How did we get our idea?

We got our idea after watching the 2023 movie 'Tetris' and thought that students should also try playing such an amazing game, which led us to come up with the idea for our project.

How to play?

When playing Tetris, you move and rotate various shapes of blocks falling from above to the left and right, and by completely filling a row, that row disappears and points are added. This is repeated until you lose.

How to use

To use our project power the Arduino that already has been coded and the game will start running.

Goals

We will introduce the game of Tetris to future children, help them understand its interesting rules, speed, and features that require logical thinking, thus assist in developing attention, problem-solving skills, and strive to promote it more widely.

Код

Функц

Манай код нь C++ хэлээр хийгдсэн ба 596 мөрнөөс тогтдог. Тухайн код нь 19 функцээс тогтдог. Үүнд:

1. bool getShape(uint8_t piece, uint8_t rot, uint8_t row, uint8_t col) - туслагч функц
2. int getLEDIndex(int row, int col) - туслагч функц
3. void playTone(int frequency, int duration)
4. void playMove()
5. void playRotate()
6. void playLineClear()
7. void playGameOver()
8. void updateLCD()
9. void showStartScreen()
10. void showGameOver()
11. void spawnPiece()
12. void rotatePiece(bool clockwise)
13. void hardDrop()
14. void clearLines()
15. void render()
16. void handleJoystick()
17. void handleButtons()
18. void restartGame()
19. void setup()

Код

```
// I piece
{
  {{0,0,0,0},{1,1,1,1},{0,0,0,0},{0,0,0,0}},
  {{0,0,1,0},{0,0,1,0},{0,0,1,0},{0,0,1,0}},
  {{0,0,0,0},{0,0,0,0},{1,1,1,1},{0,0,0,0}},
  {{0,1,0,0},{0,1,0,0},{0,1,0,0},{0,1,0,0}}
},
// O piece
{
  {{0,1,1,0},{0,1,1,0},{0,0,0,0},{0,0,0,0}},
  {{0,1,1,0},{0,1,1,0},{0,0,0,0},{0,0,0,0}},
  {{0,1,1,0},{0,1,1,0},{0,0,0,0},{0,0,0,0}},
  {{0,1,1,0},{0,1,1,0},{0,0,0,0},{0,0,0,0}}
},
// T piece
{
  {{0,1,0,0},{1,1,1,0},{0,0,0,0},{0,0,0,0}},
  {{0,1,0,0},{0,1,1,0},{0,1,0,0},{0,0,0,0}},
  {{0,0,0,0},{1,1,1,0},{0,1,0,0},{0,0,0,0}},
  {{0,1,0,0},{1,1,0,0},{0,1,0,0},{0,0,0,0}}
},
// S piece
{
  {{0,1,1,0},{1,1,0,0},{0,0,0,0},{0,0,0,0}},
  {{0,1,0,0},{0,1,1,0},{0,0,1,0},{0,0,0,0}},
  {{0,0,0,0},{0,1,1,0},{1,1,0,0},{0,0,0,0}},
  {{1,0,0,0},{1,1,0,0},{0,1,0,0},{0,0,0,0}}
},
// Z piece
{
  {{1,1,0,0},{0,1,1,0},{0,0,0,0},{0,0,0,0}},
  {{0,0,1,0},{0,1,1,0},{0,1,0,0},{0,0,0,0}},
  {{0,0,0,0},{1,1,0,0},{0,1,1,0},{0,0,0,0}},
  {{0,1,0,0},{1,1,0,0},{1,0,0,0},{0,0,0,0}}
},
// J piece
{
  {{1,0,0,0},{1,1,1,0},{0,0,0,0},{0,0,0,0}},
  {{0,1,1,0},{0,1,0,0},{0,1,0,0},{0,0,0,0}},
  {{0,0,0,0},{1,1,1,0},{0,0,1,0},{0,0,0,0}},
  {{0,1,0,0},{0,1,0,0},{1,1,0,0},{0,0,0,0}}
},
```

```
  // L piece
  {
    {{0,0,1,0},{1,1,1,0},{0,0,0,0},{0,0,0,0}},
    {{0,1,0,0},{0,1,0,0},{0,1,1,0},{0,0,0,0}},
    {{0,0,0,0},{1,1,1,0},{1,0,0,0},{0,0,0,0}},
    {{1,1,0,0},{0,1,0,0},{0,1,0,0},{0,0,0,0}}
  }
};

// Helper function to read shape from PROGMEM
bool getShape(uint8_t piece, uint8_t rot, uint8_t row, uint8_t col) {
  return pgm_read_byte(&shapes[piece][rot][row][col]);
}

// Piece Colors
const CRGB colors[7] = {
  CRGB(0, 255, 255),   // I - Cyan
  CRGB(255, 255, 0),   // O - Yellow
  CRGB(160, 0, 255),   // T - Purple
  CRGB(0, 255, 0),     // S - Green
  CRGB(255, 0, 0),     // Z - Red
  CRGB(0, 0, 255),     // J - Blue
  CRGB(255, 136, 0)    // L - Orange
};
```

```
// ===== GAME LOGIC =====

void spawnPiece() {
  if (nextType == 0) {
    currentType = random(7);
    nextType = random(7);
  } else {
    currentType = nextType;
    nextType = random(7);
  }

  currentRotation = 0;
  currentX = COLS / 2 - 2;
  currentY = 0;
}

bool checkCollision(int x, int y, uint8_t type, uint8_t rotation) {
  for (int row = 0; row < 4; row++) {
    for (int col = 0; col < 4; col++) {
      if (getShape(type, rotation, row, col)) {
        int newX = x + col;
        int newY = y + row;

        if (newX < 0 || newX >= COLS || newY >= ROWS) {
          return true;
        }

        if (newY >= 0 && grid[newY][newX] != 0) {
          return true;
        }
      }
    }
  }
  return false;
}

bool movePiece(int dx, int dy) {
  int newX = currentX + dx;
  int newY = currentY + dy;

  if (!checkCollision(newX, newY, currentType, currentRotation)) {
    currentX = newX;
    currentY = newY;
    if (dx != 0 || dy != 0) playMove();
    return true;
  }
```

```
  return false;
}

void rotatePiece(bool clockwise) {
  uint8_t newRotation = clockwise ? (currentRotation + 1) % 4 : (currentRotation + 3) % 4;

  if (!checkCollision(currentX, currentY, currentType, newRotation)) {
    currentRotation = newRotation;
    playRotate();
  } else {
    // Wall kick attempts
    int kicks[][2] = {{1, 0}, {-1, 0}, {2, 0}, {-2, 0}, {0, -1}};
    for (int i = 0; i < 5; i++) {
      if (!checkCollision(currentX + kicks[i][0], currentY + kicks[i][1], currentType, new
        currentX += kicks[i][0];
        currentY += kicks[i][1];
        currentRotation = newRotation;
        playRotate();
        break;
      }
    }
  }
}

void hardDrop() {
  while (movePiece(0, 1)) {
    score += 2;
  }
}

void lockPiece() {
  for (int row = 0; row < 4; row++) {
    for (int col = 0; col < 4; col++) {
      if (getShape(currentType, currentRotation, row, col)) {
        int y = currentY + row;
        int x = currentX + col;
        if (y >= 0 && y < ROWS && x >= 0 && x < COLS) {
          grid[y][x] = currentType + 1;
        }
      }
    }
  }
}
```

```
void render() {
  // Clear all LEDs
  FastLED.clear();

  // Draw locked pieces
  for (int row = 0; row < ROWS; row++) {
    for (int col = 0; col < COLS; col++) {
      if (grid[row][col] != 0) {
        int ledIndex = getLEDIndex(row, col);
        if (ledIndex >= 0) {
          leds[ledIndex] = colors[grid[row][col] - 1];
        }
      }
    }
  }

  // Draw ghost piece (landing preview)
  int ghostY = currentY;
  while (!checkCollision(currentX, ghostY + 1, currentType, currentRotation)) {
    ghostY++;
  }

  for (int row = 0; row < 4; row++) {
    for (int col = 0; col < 4; col++) {
      if (getShape(currentType, currentRotation, row, col)) {
        int y = ghostY + row;
        int x = currentX + col;
        if (y >= 0 && y < ROWS && x >= 0 && x < COLS && grid[y][x] == 0) {
          int ledIndex = getLEDIndex(y, x);
          if (ledIndex >= 0) {
            leds[ledIndex] = colors[currentType];
            leds[ledIndex].fadeToBlackBy(200);  // Dim ghost
          }
        }
      }
    }
  }
```

```
void clearLines() {
  int linesCleared = 0;

  for (int row = ROWS - 1; row >= 0; row--) {
    bool fullRow = true;
    for (int col = 0; col < COLS; col++) {
      if (grid[row][col] == 0) {
        fullRow = false;
        break;
      }
    }

    if (fullRow) {
      linesCleared++;

      // Shift rows down
      for (int r = row; r > 0; r--) {
        for (int c = 0; c < COLS; c++) {
          grid[r][c] = grid[r-1][c];
        }
      }

      // Clear top row
      for (int c = 0; c < COLS; c++) {
        grid[0][c] = 0;
      }

      row++;  // Check this row again
    }
  }

  if (linesCleared > 0) {
    lines += linesCleared;
    int points[] = {0, 100, 300, 500, 800};
    score += points[linesCleared] * level;

    level = lines / 10 + 1;
    moveDelay = max(100, 1000 - (level - 1) * 100);

    playLineClear();
    updateLCD();
  }
}
```

```
// ===== MAIN LOOP =====
void loop() {
  // Test LED blink
  if (millis() - lastBlinkTime > 500) {
    blinkState = !blinkState;
    digitalWrite(TEST_LED_PIN, blinkState);
    lastBlinkTime = millis();
  }

  // Check START button
  if (!gameStarted && digitalRead(BTN_START) == LOW) {
    restartGame();
    delay(200);
    return;
  }

  // Game over state
  if (gameOver) {
    showGameOver();
    playGameOver();

    // Flash red
    static unsigned long lastFlash = 0;
    static bool flashState = false;

    if (millis() - lastFlash > 500) {
      flashState = !flashState;
      lastFlash = millis();

      if (flashState) {
        fill_solid(leds, NUM_LEDS, CRGB::Red);
      } else {
        FastLED.clear();
      }
      FastLED.show();
    }

    // Wait for START to restart
    if (digitalRead(BTN_START) == LOW) {
      delay(200);
      restartGame();
    }
    return;
  }
```

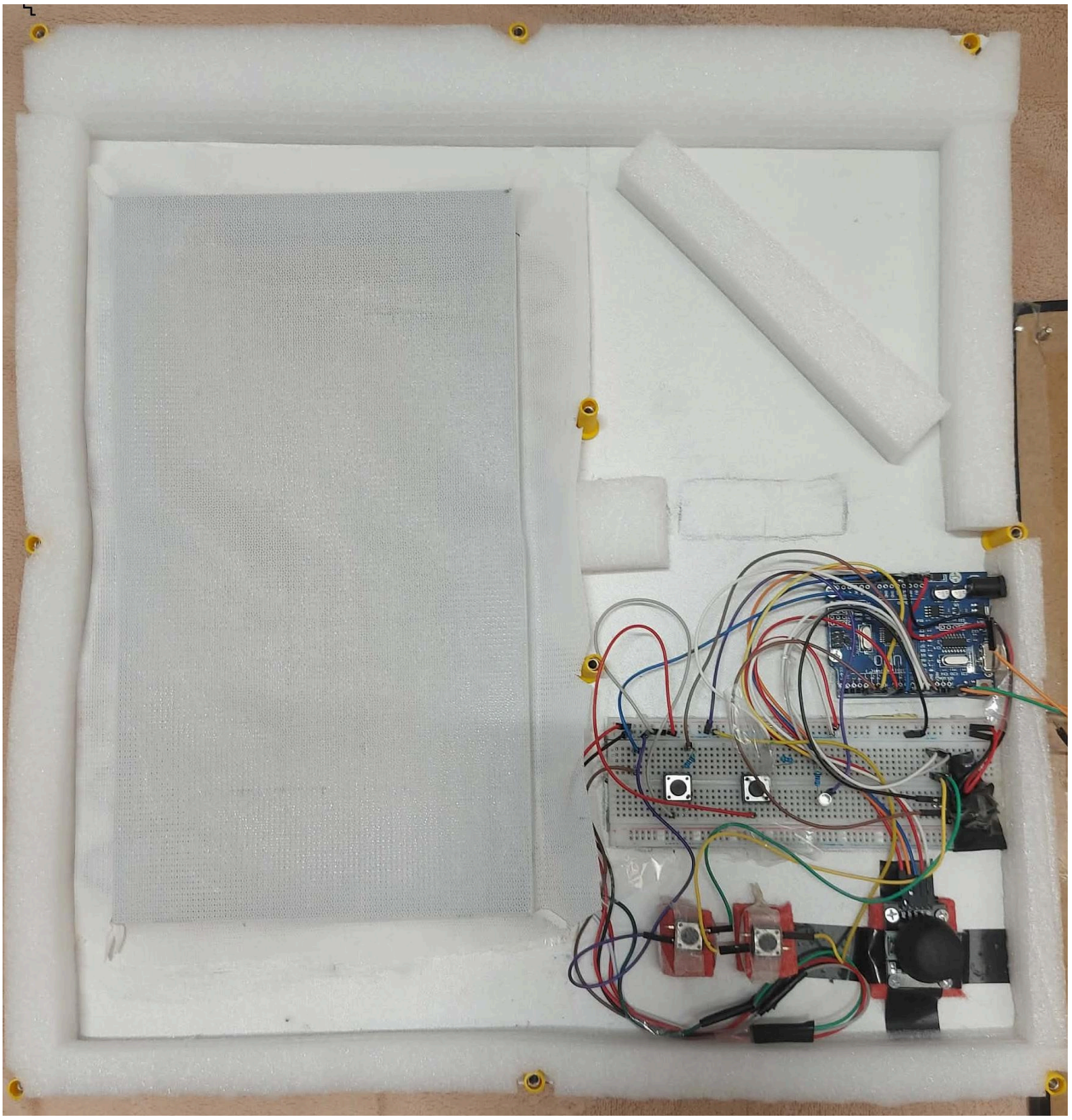
```
  // Not started yet
  if (!gameStarted) {
    playTetrisTheme();
    return;
  }

  // Paused
  if (gamePaused) {
    handleButtons();  // Only check pause button
    return;
  }

  // Normal gameplay
  handleJoystick();
  handleButtons();

  // Auto-drop piece
  if (millis() - lastMoveTime > moveDelay) {
    if (!movePiece(0, 1)) {
      lockPiece();
      clearLines();
      spawnPiece();
      if (checkCollision(currentX, currentY, currentType, currentRotation)) {
        gameOver = true;
      }
    }
    lastMoveTime = millis();
  }

  render();
}
```

ТӨСӨЛ БҮТЭЭХ ҮЙЛ ЯВЦ

Схем /Ардуино, Эд ангиуд, нийт холбол

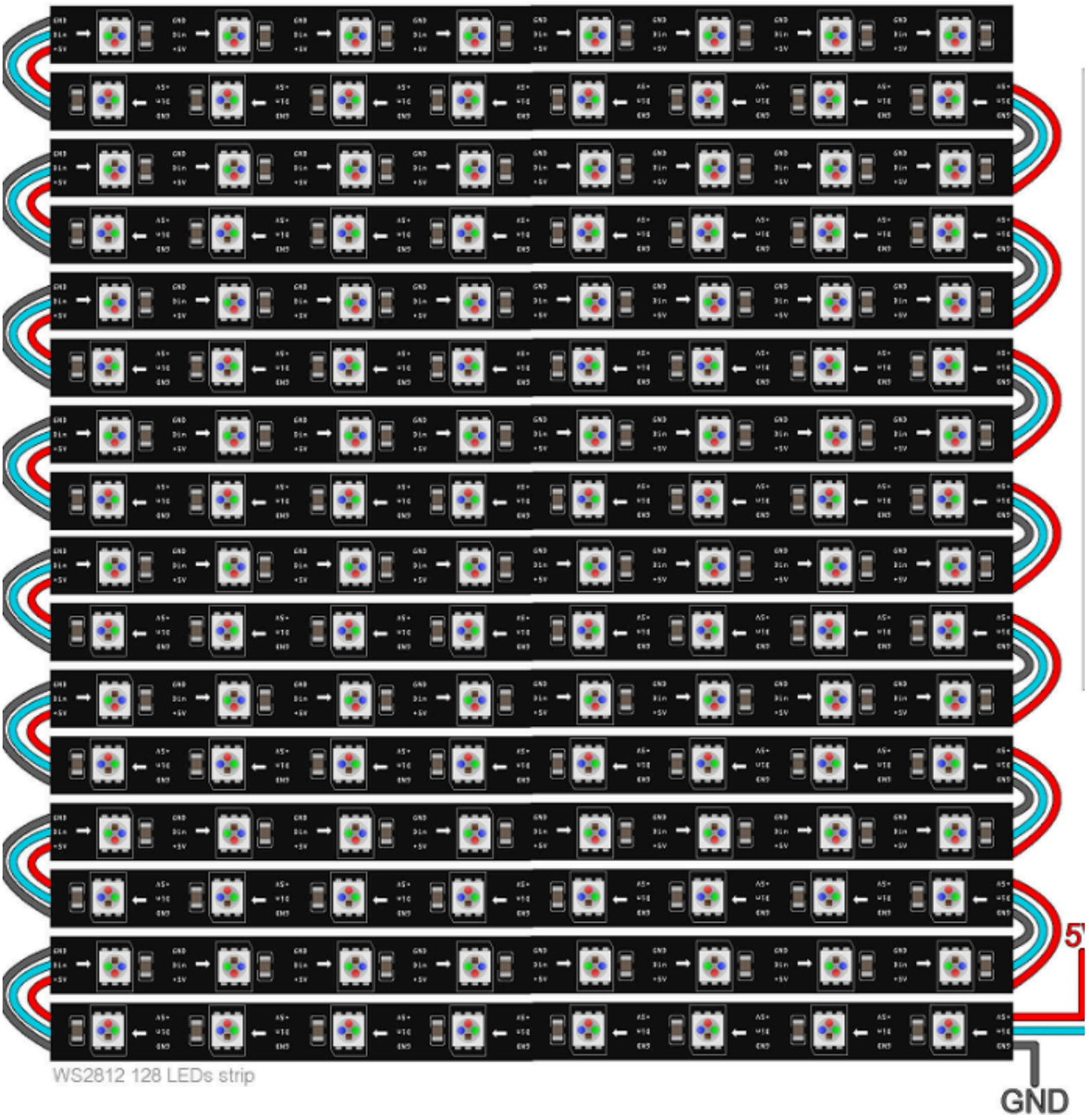
1.0

*The **outer** appearance of our project*



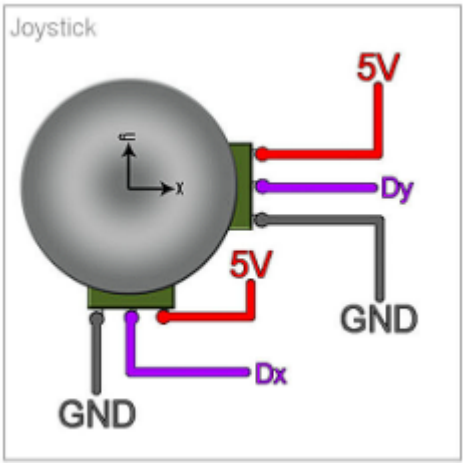
1.1

*The **inner** appearance of our project*



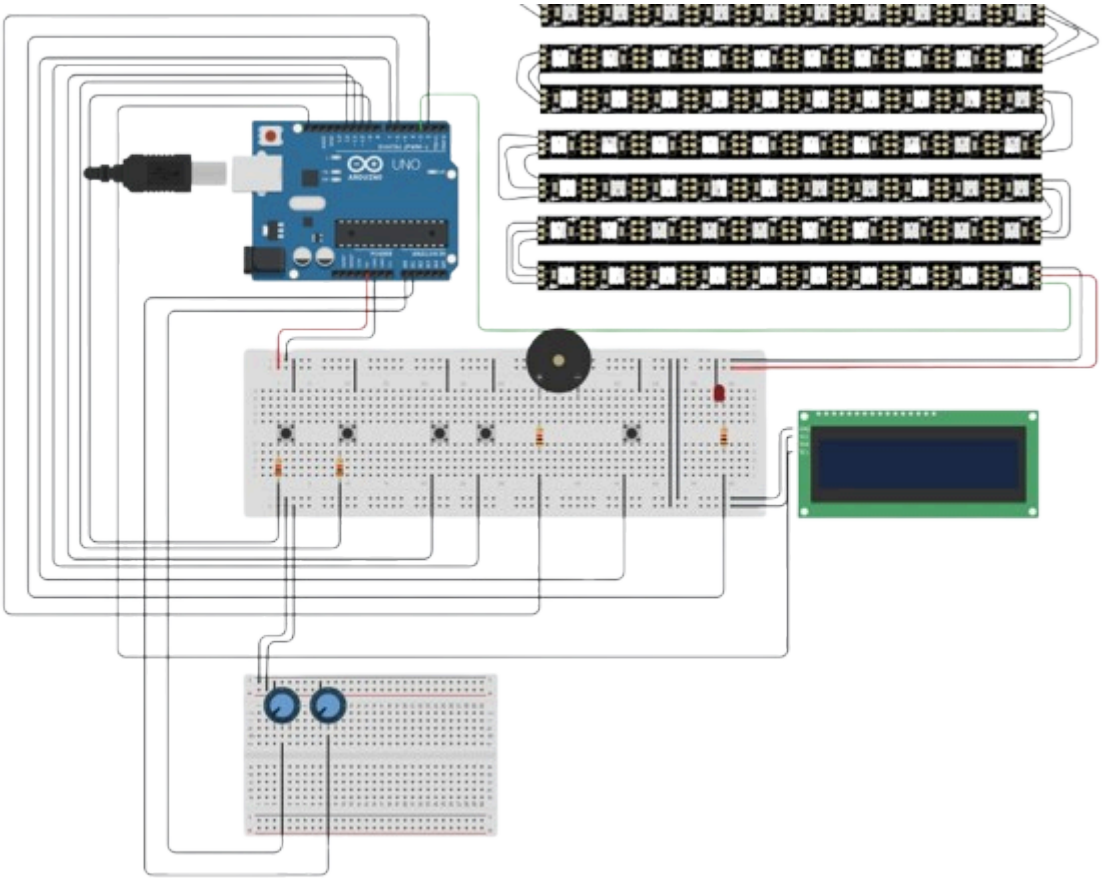
1.2

*You can see how our **LED** connect from this picture.*



1.3

*You can see how our **Joystick** connects from this picture .*



1.4

*This is an image of our project's **wiring**.*

PROJECT TESTS

Voltage power

Since our project uses Arduino it will have 5 volts of power.

Observation, Environment dependent change

From our observation if the volt is too low it will not have enough energy to light up all of the **LED** limiting sight.

USED MATERIALS

- Wooden board, Screw, and Bolt
- Arduino UNO
- Wires
- I2C display /LCD/
- Piezo
- LED
- Button, Joystick
- Bread board
- 3D Printed table like object
- Air conditioner net

PROJECT CONCLUSION

Conclusion

By implementing this project, we strengthened our team's cohesion and cooperation, improved our ability to consciously take on responsibilities, and noticeably enhanced our management together.

Judge

_____

_____